INCH-POUND
J-W-1177/42
June 10, 1988

FEDERAL SPECIFICATION SHEET

WIRE, MAGNET, ELECTRICAL, CLASS 155, TYPE SPUN, SOLDERABLE POLYURETHANE OVERCOATED WITH POLYAMIDE, ROUND

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

The requirements for acquiring the wire described herein shall consist of this specification and the latest issue of J-W-1177.

Classification:	Class 155; type SPUN (single), type SPUN2 (heavy); round.
Insulating materials:	The conductor shall be coated with a dual film. The underlying coating is based on a solderable polyurethane resin. The superimposed coating shall be based on polyamide resin.
NEMA/ANSI equivalent:	All test requirements are equivalent to MW-80 of NEMA MW 1000.
General requirements:	See J-W-1177 for general requirements, quality assurance provisions, and packaging.

Requirements:

Characteristics	Test procedure, see J-W-1177	Wire sizes, AWG	Requirements
Dimensions	4.7.1.2	14-44	See table I.
Adherence and flexibility	4.7.2.1	14-44	No cracks visible in the film coating.
Elongation	4.7.5	14-44	Not less than the value in table II.
Heat shock	4.7.4	14-44	No cracks visible in the coating after conditioning as shown in table III.

AMSC N/A FSC 6145

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Requirements: (Continued)

Characteristics	Test procedure, see J-W-1177	Wire sizes, AWG	Requirements
Scrape resistance	4.7.6	14-30	Lowest grams-to-fail load for any of the three tests and the average of the three tests shall be not less than the values in table IV.
Springback	4.7.7	14-30	Not greater than the value in table V.
Dielectric strength	4.7.9	14-44	Not less than the value in table VI.
Continuity	4.7.10	31-44	The number of discontinuities shall be not greater than the number listed in table VII.
	4.7.11	14-30	
Thermoplastic flow	4.7.8	18, 36	Median not less than 200°C with heavy film coated wire.
Solubility	4.7.12	18, 36	Heavy film coated wire shall not soften sufficiently to expose bare conductor when immersed in xylene.
Dielectric strength at temperature	4.7.14	18, 36	Heavy film coated wire shall average not less than 3825 volts for 18 AWG or 1725 volts for 36 AWG.
Thermal endurance	4.7.15.1	18	155°C minimum with heavy film coated wire.
	4.7.15.2	25-44	1000 volts/mil minimum after 168 hours at 200°C.
	4.7.15.3	25-44	175°C minimum.
Solderability	4.7.17	25-44	Covered with continuous film of solder and not readily separable after soldering as shown in table VIII.

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TABLE I. Dimensions.

AWG size	Bare wire diameter, inch			Type SPUN		Type SPUN2	
	Minimum	Nominal	Maximum	Minimum increase in diameter, inch	Maximum overall diameter, inch	Minimum increase in diameter, inch	Maximum overall diameter, inch
14	0.0635	0.0641	1/0.0644	0.0016	0.0666	0.0032	0.0682
15	.0565	.0571	1/ .0574	.0015	.0594	.0030	.0609
16	.0503	.0508	1/ .0511	.0014	.0531	.0029	.0545
17	.0448	.0453	1/ .0455	.0014	.0475	.0028	.0488
18	.0399	.0403	1/ .0405	.0013	.0424	.0026	.0437
19	.0355	.0359	1/ .0361	.0012	.0379	.0025	.0391
20	.0317	.0320	1/ .0322	.0012	.0339	.0023	.0351
21	.0282	.0285	1/ .0286	.0011	.0303	.0022	.0314
22	.0250	.0253	1/ .0254	.0011	.0270	.0021	.0281
23	.0224	.0226	1/ .0227	.0010	.0243	.0020	.0253
24	.0199	.0201	1/ .0202	.0010	.0217	.0019	.0227
25	.0177	.0179	1/ .0180	.0009	.0194	.0018	.0203
26	.0157	.0159	1/ .0160	.0009	.0173	.0017	.0182
27	.0141	.0142	.0143	.0008	.0156	.0016	.0164
28	.0125	.0126	.0127	.0008	.0140	.0016	.0147
29	.0112	.0113	.0114	.0007	.0126	.0015	.0133
30	.0099	.0100	.0101	.0007	.0112	.0014	.0119
31	.0088	.0089	.0090	.0006	.0100	.0013	.0108
32	.0079	.0080	.0081	.0006	.0091	.0012	.0098
33	.0070	.0071	.0072	.0005	.0081	.0011	.0088
34	.0062	.0063	.0064	.0005	.0072	.0010	.0078
35	.0055	.0056	.0057	.0004	.0064	.0009	.0070
36	.0049	.0050	.0051	.0004	.0058	.0008	.0063
37	.0044	.0045	.0046	.0003	.0052	.0008	.0057
38	.0039	.0040	.0041	.0003	.0047	.0007	.0051
39	.0034	.0035	.0036	.0002	.0041	.0006	.0045
40	.0030	.0031	.0032	.0002	.0037	.0006	.0040
41	.0027	.0028	.0029	.0002	.0033	.0005	.0036
42	.0024	.0025	.0026	.0002	.0030	.0004	.0032
43	.0021	.0022	.0023	.0002	.0026	.0004	.0029
44	.0019	.0020	.0021	.0001	.0024	.0004	.0027

1/ The maximum bare wire dimensions may be exceeded up to the NEMA/ANSI maximum bare wire limit, provided the minimum increase is maintained and the maximum overall diameter specified is not exceeded.

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TABLE II. Elongation.

AWG size	Elongation, minimum percent
14	33
15	33
16	32
17	32
18	32
19	31
20	30
21	30
22	29
23	29
24	28
25	28
26	27
27	27
28	26
29	26
30	25
31	24
32	24
33	23
34	22
35	21
36	20
37	20
38	19
39	18
40	17
41	17
42	16
43	15
44	14

TABLE III. Heat shock.

AWG size	Minimum elongation, percent	Mandrel diameter	Minimum temperature, °C
14-30	20	3X	175
31-44	1/20	3X	175

1/ Or to the breaking point, whichever is less.

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TABLE IV. Scrape resistance.

AWG size	Type SPUN		Type SPUN2	
	Scrape, grams-to-fail		Scrape, grams-to-fail	
	Average	Minimum	Average	Minimum
14	840	715	1490	1270
15	780	665	1400	1190
16	735	625	1310	1115
17	690	585	1200	1045
18	645	550	1150	950
19	600	510	1070	910
20	560	475	1000	850
21	525	445	940	800
22	490	415	880	750
23	460	390	820	700
24	430	365	770	655
25	400	340	720	615
26	380	325	675	575
27	355	300	635	540
28	335	285	595	510
29	310	265	560	480
30	295	250	525	450

TABLE V. Springback.

AWG size	Springback, maximum degrees per turn
14	42
15	46
16	50
17	54
18	58
19	62
20	66
21	53
22	58
23	62
24	67
25	72
26	76
27	50
28	55
29	61
30	66

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TABLE VI. Dielectric strength.

AWG size	Type SPUN	Type SPUN2
	Dielectric strength, minimum breakdown volts	Dielectric strength, minimum breakdown volts
14	3175	5700
15	3075	5550
16	3000	5400
17	2925	5275
18	2850	5125
19	2775	5000
20	2700	4650
21	2625	4725
22	2575	4625
23	2500	4500
24	2425	4375
25	2375	4250
26	2300	4150
27	2250	4050
28	2175	3950
29	2150	3825
30	2075	3725
31	1875	3450
32	1675	3175
33	1500	2925
34	1350	2675
35	1200	2475
36	1075	2275
37	975	2100
38	850	1925
39	775	1775
40	700	1625
41	625	1500
42	575	1375
43	500	1250
44	450	1175

TABLE VII. Continuity.

AWG size	Maximum number of discontinuities	
	Type SPUN	Type SPUN2
14-24	25	5
25-30	25	7
31-44	25	5

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TABLE VIII. Solderability.

AWG size	Maximum immersion time, seconds		Temperature of solder, °C
	Type SPUN	Type SPUN2	
14-19	10	10	430
20-23	8	8	430
24-29	6	6	360
30-36	5	5	360
37-44	4	4	360

Part number: Magnet wire covered by this specification shall be defined by the following part numbering system. Example: M1177/42-02C029.

M1177/42-	02	C	029
Federal specification identifier	Two digit type code	Single letter conductor code	Three character size code

The following codes shall apply:

Type	Type code	Conductor	Conductor code
SPUN	01	Copper	C
SPUN2	02	Aluminum	A
		Nickel-coated copper	N
		Silver-coated copper	S

The size code shall be the bare wire dimension. AWG wire size shall be used.

Intended use: Type SPUN magnet wire is intended for use in 155°C applications where solderability and good windability is desired.

MILITARY INTERESTS:

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - AR, ER, MI
DLA - IS

User activities:
Army - ME
Navy - AS, CG, MC, OS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS, PBO, PCD
INTERIOR - BLM
HHS - FDA
DCGOVT - DCG
NASA - JFK
COMMERCE - NBS
TRANSPORTATION - APM, FAA

Preparing activity:
Navy - SH
(Project 6145-1111-38)

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